

Leaky Integrate-and-Fire (LIF)

Part I: Overview

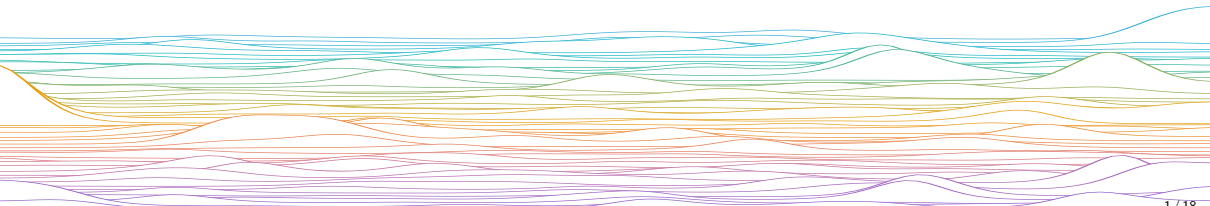
Computational Neuroscience

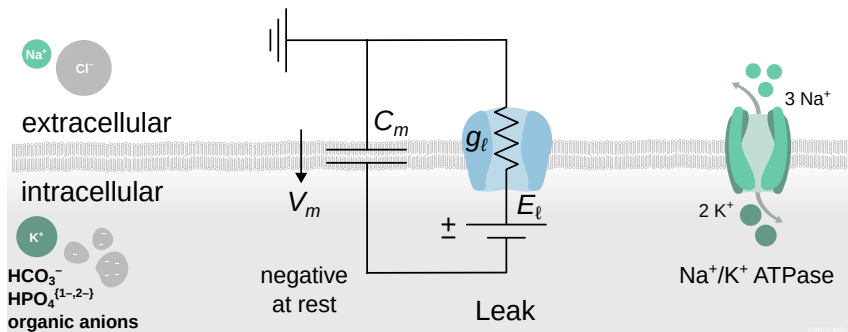
University of Bristol

M Rule

Learning outcomes:

- ▶ Mathematically analyze spiking dynamics of the LIF model
- ▶ The $f-I$ curve (rate model) for LIF





$$C_m \dot{v} = g_\ell (E_\ell - v)$$

$$\tau_m \dot{v} = E_\ell - v$$

$$C_m = 100 \times 10^{-12} \text{ Farads} \quad (100 \text{ picofarads})$$

$$E_\ell = -70 \times 10^{-3} \text{ Volts} \quad (-70 \text{ millivolts})$$

$$g_\ell = 10 \times 10^{-9} \text{ Siemens} \quad (10 \text{ nanosiemens})$$

$$R_m = 1/g_\ell = 100 \times 10^6 \text{ Ohms} \quad (100 \text{ megaohms})$$

$$\tau_m = r_m C_m = 10 \times 10^{-3} \text{ seconds} \quad (10 \text{ milliseconds})$$

cell membrane capacitance

resting potential

leak conductance

membrane resistance

membrane time constant

Modelling Scales

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin–Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch–Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability

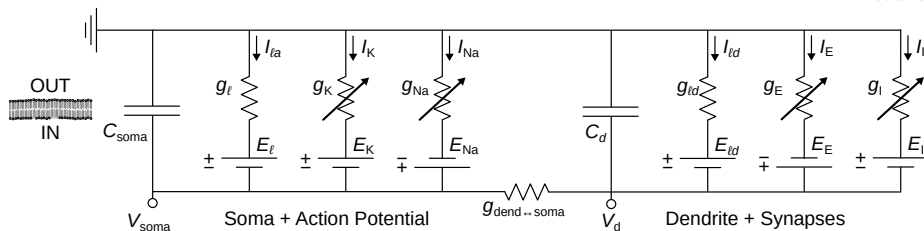
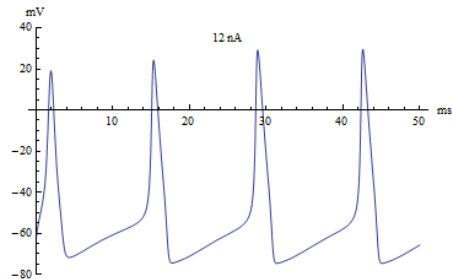


Phenomenological,
Qualitative

Conductance Model Point Neuron e.g. Hodgkin–Huxley

Nonlinear, action potential costly to simulate

Study role of ion channels, conductances, dendritic morphology (shape), etc.

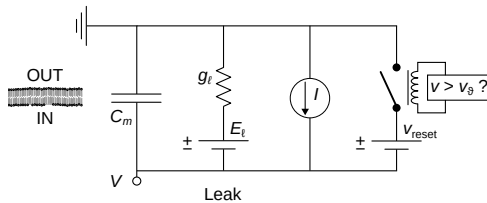
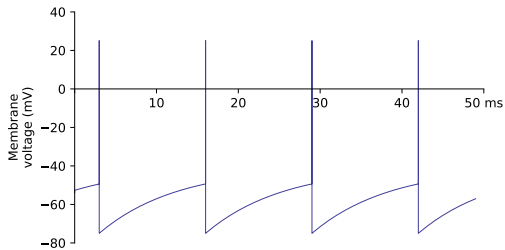


do you need to fix any mistakes in the diagram?

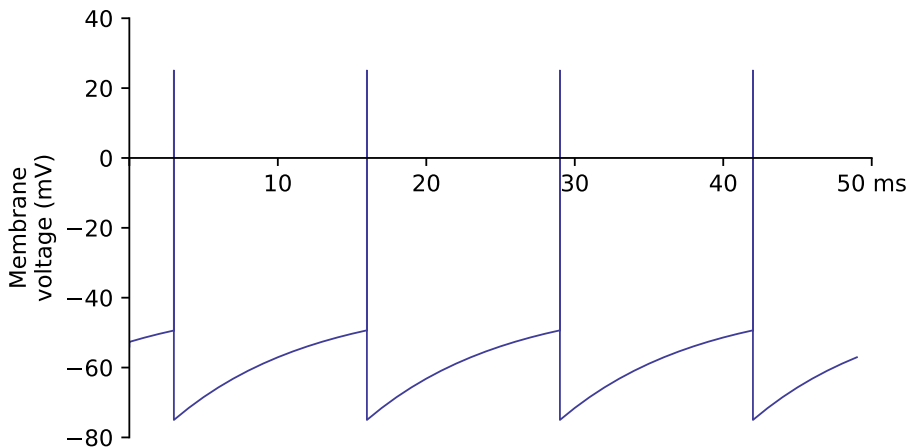
Leaky Integrate-and-Fire (LIF)

Departs from physiology, but
sufficient to build intuition

Easy to integrate



Leaky Integrate-and-Fire (LIF)



LIF model:

$$C\dot{v} = \frac{1}{R}(v_r - v) + I$$

if $v(t^-) > v_\theta$ then

Emit a spike

$v(t)$: membrane voltage

v_θ : Threshold (spike when $v(t) > v_\theta$)

v_r : Reset voltage ($v(t) \leftarrow v_r$ after spike)

R : Membrane resistance

Solution of LIF up until spike?